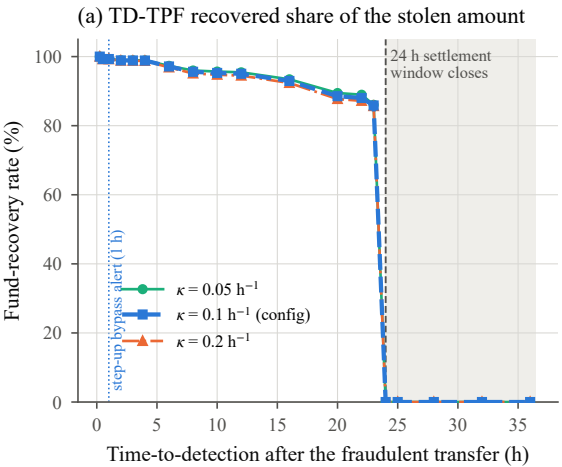
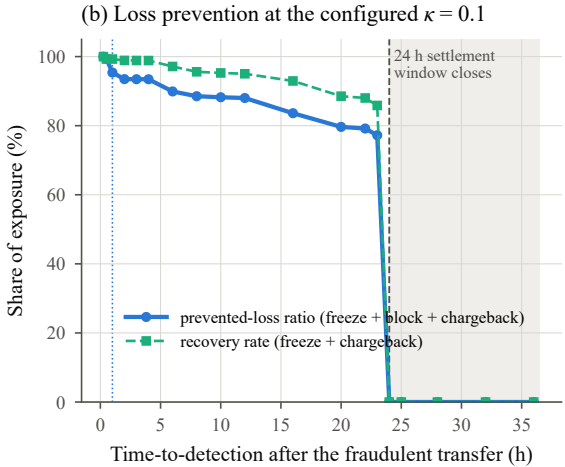

(a) TD-TPF recovered share of the stolen amount
Fund-recovery rate (%)
100
80
60
40
20
0
24 h settlement window closes
step-up bypass alert (1 h)
κ = 0.05 h⁻¹
κ = 0.1 h⁻¹ (config)
κ = 0.2 h⁻¹
0
5
10
15
20
25
30
35
Time-to-detection after the fraudulent transfer (h)


(b) Loss prevention at the configured κ = 0.1
Share of exposure (%)
100
80
60
40
20
0
24 h settlement window closes
prevented-loss ratio (freeze + block + chargeback)
recovery rate (freeze + chargeback)
0
5
10
15
20
25
30
35
Time-to-detection after the fraudulent transfer (h)